

Quiz 1

Each question the same marks. Justify your answers.

Q 1

Given a, b are integers what is the smallest positive value D of the determinant of the matrix

$$A = \begin{pmatrix} 38 & 60 & c \\ a & b & d \\ 0 & 0 & 1 \end{pmatrix}$$

Exhibit a, b, c and d such that the matrix has determinant D .

Answer 1

We know that $g = \gcd(38, 60) = 2$. Since g divides $\det(A) = 38b - 60a$, we see that the smallest possible value of this determinant is 2.

We use the GCD algorithm to find that $2 = 7 \cdot 60 - 11 \cdot 38$. Thus we can take

$$(a, b, c, d) = (-7, -11, 0, 0)$$

Non-quiz

Consider the sequence f_n defined by $f_0 = 0, f_1 = 1$ and

$$f_{n+2} = af_n + bf_{n+1}; \text{ for } n \geq 0$$

Under what conditions does f_n grow exponentially?

Answer

By the standard solution of the quadratic equation $T^2 = a + bT$, we note that

$$\alpha = \frac{b + \sqrt{b^2 + 4a}}{2}$$

satisfies $\alpha^2 = a + b\alpha$. The other solution of this equation is $b - \alpha$.

Note that both of these solutions satisfy the equations $T^{n+2} = aT^n + bT^{n+1}$. It follows that if we write $g_n = c\alpha^n + (-1)^n d(\alpha - b)^n$, then

$$g_{n+2} = ag_n + bg_{n+1}$$

Thus if we want $f_n = g_n$, we need only solve

$$\begin{aligned} c + d &= 0 \\ c\alpha - d(\alpha - b) &= 1 \end{aligned}$$

for c and d . We write $d = -c$ to solve the first equation. The second equation then becomes

$$c(2\alpha - b) = 1$$

which has a solution provided $2\alpha \neq b$. Note that $2\alpha = b$ if and only if the discriminant $b^2 + 4a$ of the above quadratic equation is 0. In particular, if $a > 0$, then this discriminant is non-zero.

In that case, we obtain the formula

$$f_n = \frac{\alpha^n - (-1)^n(\alpha - b)^n}{2\alpha - b}$$

Note that α and $\alpha - b$ are positive if $a > 0$ and $\alpha > \alpha - b$ if $b > 0$. Moreover, in this case $2\alpha > b$. Furthermore, if $a + b > 1$, then $\alpha > 1$. Thus, under these assumptions, the term $\alpha^n/(2\alpha - b)$ dominates the growth of f_n . Hence, its growth is exponential.

If $a > 0$, $b > 0$ and $a + b > 1$, then f_n grows exponentially.